

Unit:-3

Network Layer in OSI Model

- The Network Layer is the 5th Layer from the top and the 3rd layer from the Bottom of the [OSI Model](#).
- It is one of the most important layers which plays a key role in data transmission.
- The main job of this layer is to maintain the quality of the data and pass and transmit it from its source to its destination.
- It also handles [routing](#), which means that it chooses the best path to transmit the data from the source to its destination, not just transmitting the packet.
- There are several important protocols that work in this layer.
- The network layer is also responsible for organizing and controlling the available paths for data transfer.

Functions of Network Layer

- Gives each device a unique IP address for identification and communication.
- Breaks data into smaller packets for easy transmission.
- Ensures data reaches the correct destination.
- Directs packets through routers to the right path.
- Splits large data into smaller parts for transmission and puts them back together at the destination.
- Divides a big network into smaller ones for better organization and faster communication.
- Converts private IP addresses to public ones for internet access and security.
- Finds the best route for data to travel across different networks.

How Does the Network Layer Work?

- Each device gets a unique IP address for identification.
- Data is broken into small packets with source and destination details.
- Routers decide the best route for packets to reach their destination.
- Packets move through different routers until they arrive at the correct device.
- Large packets are split into smaller parts to fit through the network.
- The original data is reassembled at the destination.
- If delivery fails, an error message is sent back.

The protocols used in the Network Layer:

1. IP (Internet Protocol)

- **What it does:** IP gives each device on a network a unique address (like an address on a letter) so that data can find its way to the right place. There are two versions:
 - **IPv4:** Older version with 32-bit addresses (e.g., 192.168.0.1).
 - **IPv6:** Newer version with 128-bit addresses (e.g., 2001:0db8:85a3...).

2. ICMP (Internet Control Message Protocol)

- **What it does:** ICMP is used for sending error messages or diagnostic info. For example, if a device can't be reached, it will send a message back saying "destination unreachable."
 - **Example:** The **Ping** command uses ICMP to check if a device is online.

3. ARP (Address Resolution Protocol)

- **What it does:** ARP helps devices find each other on the same local network. If a device knows an IP address but not the physical address (MAC address), ARP helps to map the IP to the MAC address.
 - **Example:** A computer asks, "Who has IP 192.168.0.5?" to find the device at that address.

4. RARP (Reverse Address Resolution Protocol)

- **What it does:** RARP was used to find the IP address of a device when it only knew its MAC address. It's not used much anymore because **DHCP** does this job better.

5. Routing Protocols (These help routers decide the best path for data to travel):

- ✓ Routing protocols help routers determine the best path for forwarding data packets between networks. They dynamically update routing tables and ensure efficient data delivery.
- **RIP (Routing Information Protocol):** An older protocol that helps routers share info about network paths. It uses "hop count" to find the best route.
- **OSPF (Open Shortest Path First):** A faster, better protocol than RIP. It's used in larger networks and finds the best route based on the "cost" (bandwidth).

Advantages of the Network Layer :

- Breaks data into small packets for faster and easier transfer.
- Finds the best route for data to reach its destination.
- Detects and reports errors in data transmission.
- Assigns unique IP addresses to identify devices.
- Works smoothly across different types of networks.
- Manages traffic to prevent network slowdowns.
- Sends data to multiple devices at the same time.

What is Network Switching?

- ✓ **Switching** in networking refers to the process of directing data from one device to another within a network. It determines how data is transmitted from the sender to the receiver, using specific paths.

or

- ✓ Network Switching refers to the process of directing data packets between devices in a network, typically using network devices called switches. A switch acts as a controller that connects multiple devices on the same network, like computers or printers, and helps forward data between them.

Key Concepts:

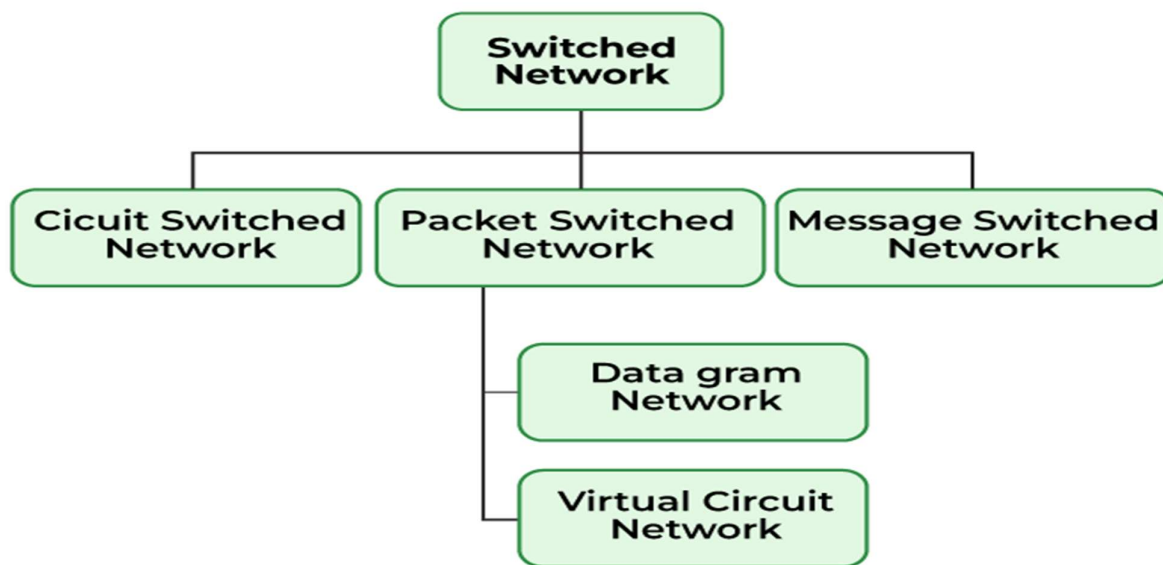
- **Learning:** Switches "learn" the addresses of connected devices and store them in a table to know where to forward the data.

- **Forwarding:** Switches forward packets to the correct port based on the MAC address (Layer 2) or IP address (Layer 3).
- **Broadcasting:** If the switch doesn't know the destination device, it sends the packet to all connected devices (broadcast) and waits for the recipient to respond.

Types of Network Switching:

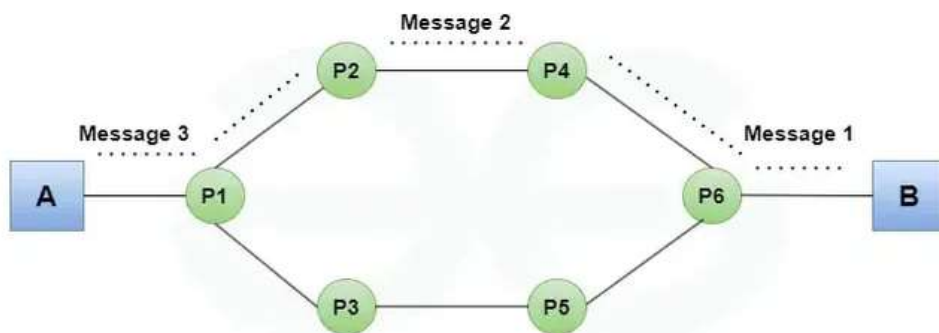
1. **Layer 2 Switching** (Data Link Layer):
 - Switches use **MAC addresses** to forward packets within the same network.
2. **Layer 3 Switching** (Network Layer):
 - Switches use **IP addresses** to route packets between different networks.

There are different types of switching :



1. Circuit Switching

- **Circuit Switching** is a method used in telecommunications to establish a dedicated communication path between two devices for the entire duration of a call or data transmission.



Example:

- **Traditional Telephone Networks:** When you make a phone call, a dedicated path is set up between your phone and the recipient's phone for the entire duration of the call.

Advantages of Circuit Switching:

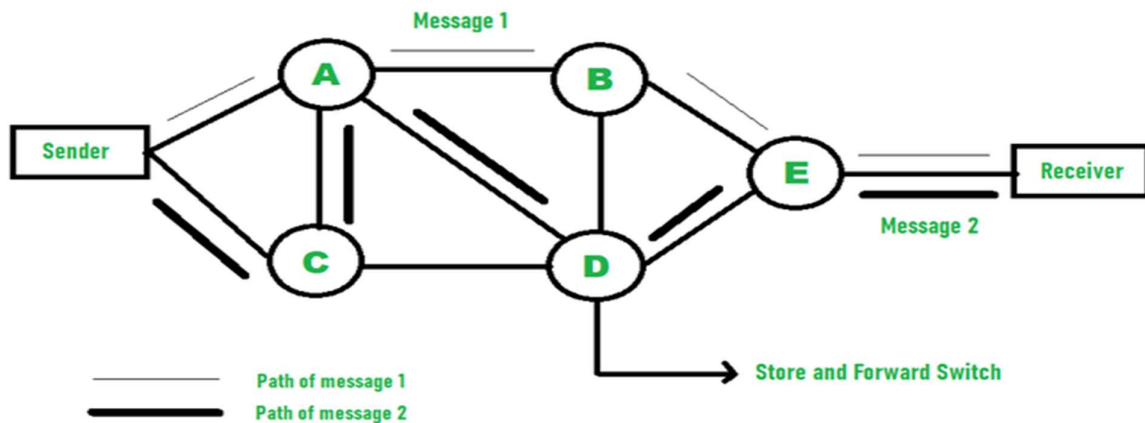
- Reliable communication with stable quality.
- No interference during the call or data transfer.
- Consistent connection for the entire duration.
- Simple to implement for voice communication

Disadvantages of Circuit Switching:

- Inefficient use of resources when the path is idle.
- Wastes bandwidth if there's no active communication.
- Not flexible for handling multiple connections at once.
- Takes time to set up the dedicated connection.
- Limited scalability for large networks.

2. Packet Switching

- **Packet Switching** is a method of data transmission where data is broken into smaller packets and each packet is sent independently across the network.
- The packets may take different paths to reach the destination, where they are reassembled into the original data.



Example:

- **The Internet:** Most internet data (emails, web pages, videos) is transmitted using packet switching.

Advantages:

- Efficient use of bandwidth.
- More flexible and scalable.
- Better suited for data traffic that's intermittent (like web browsing or emails).

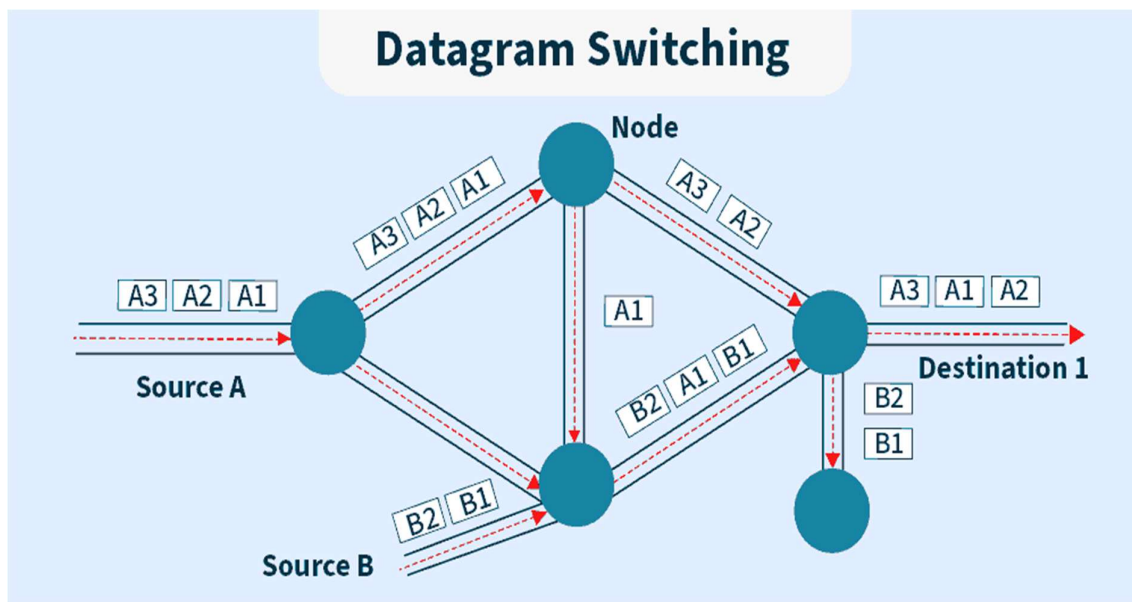
Disadvantages:

- Packets can arrive out of order or be lost.
- Slight delay may occur due to multiple routing paths.

There are **three main types of packet switching**:

1. Datagram Packet Switching:

- Each packet is sent independently.
- Packets may take different routes to reach the destination.
- No need to establish a connection beforehand.



Advantages of Datagram Packet Switching:

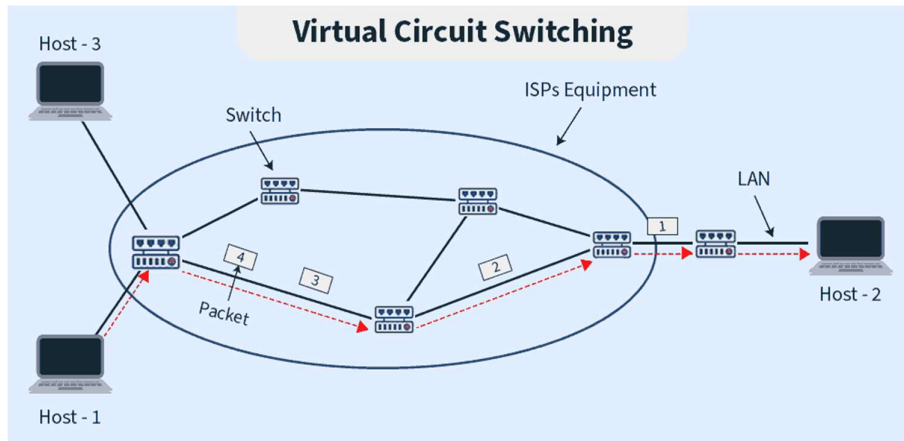
- No connection setup required.
- Flexible routing of packets.
- Efficient use of network resources.
- Scalable for large networks.

Disadvantages of Datagram Packet Switching:

- Higher overhead due to extra routing info.
- Possible delays due to varying paths.
- No Quality of Service (QoS) guarantee.

2. Virtual Circuit Packet Switching:

- ✓ In Virtual Circuit Packet Switching, a logical path (called a virtual circuit) is established between the sender and receiver before data transmission begins. Unlike Datagram switching, packets follow the same path, ensuring they arrive in order.



Key Features:

- Connection Setup:** A virtual circuit is created before data transfer, and all packets follow the same path.
- Ordered Delivery:** Since packets follow the same route, they arrive in the correct order.
- Reliability:** The network manages the connection, ensuring reliable data delivery.

Advantages of Virtual Circuit Packet Switching:

- Ordered packet delivery.
- Reliable communication.
- Suitable for real-time applications.

Disadvantages of Virtual Circuit Packet Switching:

- Requires connection setup.
- Less flexibility (affected by network failures).
- Resource-intensive for short transfers.

3. Hybrid Packet Switching:

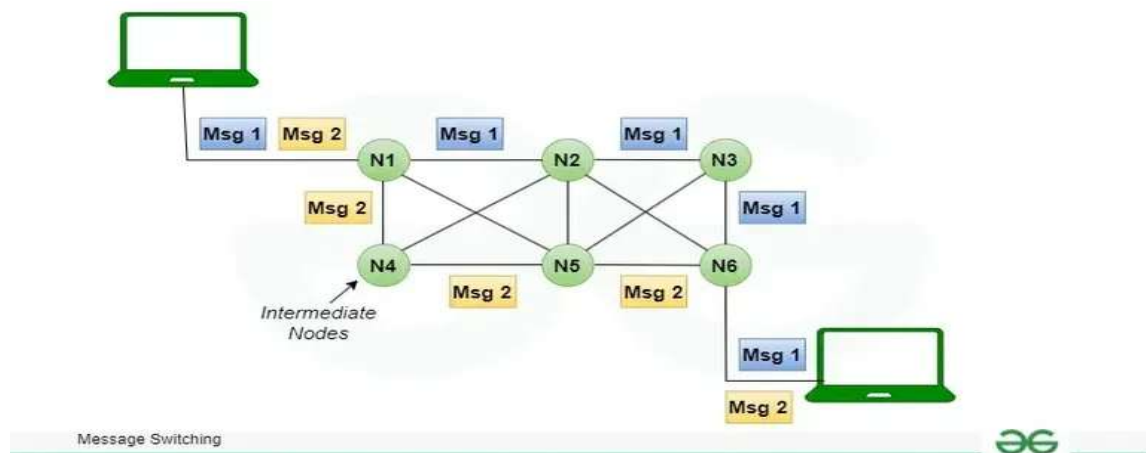
- Combines elements of both Datagram and Virtual Circuit.
- Can switch between the two types depending on the need for flexibility or reliability.

Difference Between Datagram Switching and Virtual Circuit Switching

Datagram Switching	Virtual Circuit Switching
No connection setup needed.	Requires a connection setup before data transfer.
Each packet may take a different route.	All packets follow the same pre-established path.
Unordered packets; packets can arrive out of order.	Ordered packet delivery.
Highly flexible, packets can be routed independently.	Less flexible, as the route is fixed.
Less reliable (packets may get lost or arrive out of order).	More reliable (ensures consistent delivery).
Lower overhead per packet.	Higher overhead due to connection management.

3. Message Switching :

- **Message Switching** is a method of communication where the entire message is sent as a whole from the source to the destination, but it is stored temporarily at intermediate nodes (like routers) before being forwarded to the next node.



How Message Switching Works:

- The sender sends the **entire message** to the first node.
- The message is **stored** temporarily at each node.
- The node **forwards** the message to the next node when it's ready.
- This continues until the message reaches its **final destination**.

Example:

- **Telegraph System:** In early communication systems, messages were sent, stored temporarily at intermediate stations, and then forwarded to the destination.

Advantages of Message Switching:

- Cost-effective.
- Good for long-distance communication.
- Handles large messages.
- Saves network resources.
- Reliable message delivery.

Disadvantages of Message Switching:

- Causes delays due to storing messages at each node.
- Requires a lot of storage at intermediate nodes.
- Not suitable for real-time communication.
- Risk of message loss if nodes are full or fail.
- Less efficient compared to packet switching.

Physical Addressing & Logical Addressing

❖ Physical Addressing :

- The **Physical Address** is also known as the **MAC (Media Access Control)** address. It is a unique identifier assigned to a device for communication within a network.
- The physical address is used by the **Data Link Layer** (Layer 2) and is the lowest-level address in the OSI model.
- The size of a physical address is **48 bits (6 bytes)**.

Key Points: Physical Addressing:

- ❖ Uses MAC addresses (unique to each device).
- ❖ Works at the Data Link Layer (Layer 2).
- ❖ Local to a specific network or device.
- ❖ Hardcoded into network devices

Example of Physical Addressing:

- **MAC Address Example:**

16 : 1A : BB : 6F : 90 : E5

- This is a unique identifier for a network interface card (NIC) in a device like a computer, router, or switch.

- Used for communication within the same local network.

Advantages of Physical Address (MAC Address):

- Every device has a unique built-in address.
- Helps control who can connect to the network.
- Ensures data reaches the correct device.
- No manual setup is required; it works automatically.
- Best for communication within the same local network.

Disadvantages of Physical Address (MAC Address):

- Can be faked or changed by hackers.
- Only works within the same local network.
- Difficult to modify if needed.
- Less secure in large networks.
- Cannot be used for routing data across different networks.

There are two main types of **Physical Addressing** based on the context in which they are used:

1. Unicast Addressing:

- A unique address assigned to a single device within the network.
- Example: A specific MAC address for a network interface card (NIC).

2. Multicast Addressing:

- An address used to send data to multiple devices within a network (group of devices).
- Example: A multicast MAC address used for broadcasting data to a specific group.

❖ Logical Addressing:

- **Logical Addressing** refers to the IP address used to identify devices across different networks.
- It operates at the **Network Layer (Layer 3)** of the OSI model, which is responsible for routing data packets between different networks.
- The size of a logical address (IP address) is either **32 bits** (IPv4) or **128 bits** (IPv6).

Key Points: Logical Addressing:

- **Uses IP addresses** (assigned to devices across different networks).
- **Works at the Network Layer (Layer 3).**
- **Global:** Can be used across networks, not limited to a specific one.
- **Assignable:** Can be dynamically assigned via DHCP or manually configured.

Example of Logical Addressing:

- **IPv4 Address Example: 192.168.1.1**
 - This is a unique identifier assigned to a device in an IP network.
 - It is used for communication across different networks.
- **IPv6 Address Example: 2001:0db8:85a3:0000:0000:8a2e:0370:7334**
 - This is a 128-bit address written in eight groups of four hexadecimal digits.
 - IPv6 is used for communication across networks, just like an IPv4 address, but offers a much larger address pool due to its larger size (compared to the 32-bit IPv4 address).

Advantages of Logical Addressing:

- Supports many devices, especially with IPv6.
- Allows communication between different networks.
- Can be changed without affecting the device hardware.
- Helps data travel between networks using routers.
- Enables global communication over the internet.

Disadvantages of Logical Addressing:

- Requires more processing and management than physical addressing.
- Uses extra resources for assigning, routing, and translating addresses.
- More vulnerable to attacks like IP spoofing.
- Needs proper setup of IP addresses and routing.
- Depends on routers and switches for communication between networks.

Here's the comparison between Physical Addressing and Logical Addressing:

Physical Addressing	Logical Addressing
Identifies devices at the hardware level.	Identifies devices based on their network location.
Uses MAC addresses (48 bits).	Uses IP addresses (IPv4: 32 bits, IPv6: 128 bits).
Operates at the Data Link Layer (Layer 2).	Operates at the Network Layer (Layer 3).
Works within a local network (not across networks).	Can work across different networks (internet).
Fixed address assigned by the manufacturer.	Assigned dynamically or statically in a network.
Cannot be changed or easily modified.	Can be changed or reassigned.

Provides communication in the same physical network.	Provides communication across networks.
Used for delivering data to the correct device locally.	Used to route data from one network to another.

There are two primary types of **logical addressing** used in networking:

Logical addressing is used in networking to uniquely identify devices across different networks. The two primary types are:

1. **IPv4 (Internet Protocol version 4) Addressing**
2. **IPv6 (Internet Protocol version 6) Addressing**

1. IPv4 (Internet Protocol Version 4)

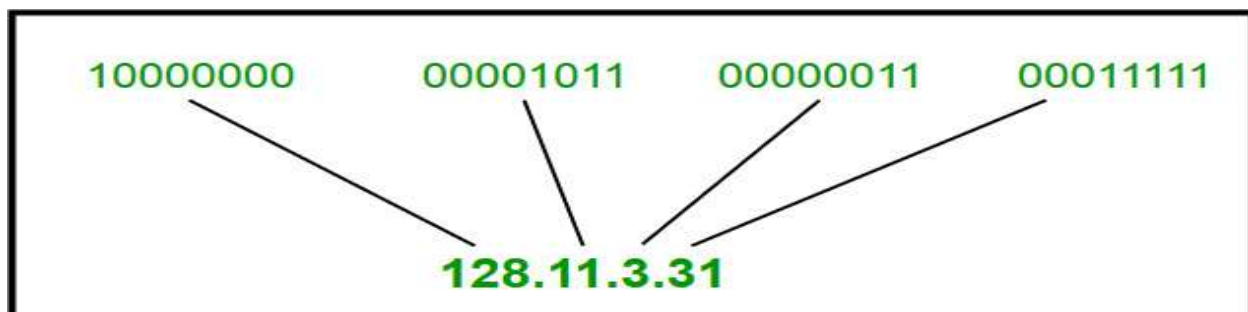
- ✓ IPv4 (Internet Protocol version 4) is the original addressing system used on the Internet,
- ✓ Introduced in **1983**, it is the **most widely used** Internet addressing system.
- ✓ It Uses a **32-bit address** (e.g., **192.168.1.1**).
- ✓ Supports **over 4 billion unique addresses** (2^{32}).
- ✓ Addresses are written in **dotted decimal notation**
- ✓ Divided into **network and host** parts.
- ✓ Uses **subnetting and CIDR** for efficient allocation.
- ✓ Used for **device communication** over networks.

Example: 128.11.3.31

- 128 is the first octet
- 11 is the second octet
- 3 is the third octet
- 31 is the fourth octet

IPv4 Address Format :

IPv4 Address Format is a 32-bit Address that comprises binary digits separated by a dot (.).



2.) IPv6 (Internet Protocol Version 6)

- **IPv6 is the successor to IPv4, designed to overcome the limitations of IPv4.**
- Uses a **128-bit address** (e.g., **2001:db8::1**).
- Supports **3.4×10^{38} unique addresses** (virtually unlimited).
- Addresses are written in hexadecimal format, typically divided into 8 groups of 4 hexadecimal digits (e.g- 2001:0db8:85a3:0000:0000:8a2e:0370:7334).
- Written in **hexadecimal notation** and separated by **colons (:: for compression)**.
- Designed to replace IPv4 due to address exhaustion.
- Supports auto-configuration, better security, and improved routing.
- Eliminates Network Address Translation (NAT), enabling direct device communication.

IPv6 provides more addresses, better security, and improved efficiency, making it the future of the internet.

Types of IPv6 Addresses

IPv6 addresses are categorized into three main types:

Unicast Address & Multicast Address & Anycast Address

1. Unicast Address

- Used for **one-to-one communication** (single sender and a single receiver.) between devices.
- Data is sent to a specific **single** recipient.
- Types of Unicast Addresses:
 - **Global Unicast:** Public, routable on the internet (like IPv4 public addresses).
 - **Link-Local:** Used for communication within the same network segment (**FE80::/10**).
 - **Unique Local:** Private address space, similar to IPv4 private addresses (**FC00::/7**).

2. Multicast Address

- Used for **one-to-many communication**.
- Packets are delivered to **multiple nodes** that have joined a multicast group.
- Replaces **broadcasting** in IPv4.
- Always starts with **FF00::/8**.

3. Anycast Address

- Used for **one-to-nearest communication**.
- Packets are sent to **the closest device** in a group of multiple devices with the same address.
- Mainly used for **load balancing and routing optimization**.
-

Benefits of IPv6 over IPv4:

- Supports a much larger number of unique addresses.
- Has a simpler and more efficient header for faster processing.
- Provides built-in security with mandatory IPsec support.
- Removes the need for NAT since there are enough unique addresses.

Advantages of IPv6

- Provides a 128-bit address space, solving IPv4 address shortage.
- Has built-in IPsec encryption for better security.
- Allows direct device communication without the need for NAT.
- Supports automatic IP configuration using SLAAC and DHCPv6.
- Improves routing and multicasting efficiency with no need for broadcasts.

Disadvantages of IPv6

- Slow adoption (Many networks still use IPv4).
- High upgrade cost (New infrastructure needed).
- Larger addresses (More memory usage).
- Security risks (New vulnerabilities).

Here's a comparison of **IPv4** and **IPv6** :

IPv4	IPv6
32-bit address length	128-bit address length
Written in decimal (e.g., 192.168.1.1)	Written in hexadecimal (e.g., 2001:0db8::85a3:0000)
~4.3 billion unique addresses	340 undecillion unique addresses
More complex header with 12 fields	Simplified header with 8 fields
Security is optional (IPSec supported)	IPSec is mandatory (built-in security)
Manual or DHCP configuration	Auto-configuration (stateless) and DHCPv6
Uses broadcasting	No broadcasting, uses multicast
Requires NAT due to limited address space	No need for NAT due to large address space
Uses classful routing and subnetting	More efficient hierarchical routing
Includes a checksum in the header	No checksum in the header
Packet fragmentation by routers and sender	Packet fragmentation only by the sender
Compatible with older devices and networks	Requires a transition from IPv4

-: Address Mapping :-

- ✓ Address mapping is the process of translating logical (IP) addresses to physical (MAC) addresses or vice versa.
- ✓ It plays a crucial role in network communication by ensuring that data packets reach the correct destination.

There are two main types of address mapping:

1. Static Mapping :

- ✓ In **static address mapping**, a manual one-to-one mapping is created between an IP address and a MAC address.

Characteristics of Static Mapping:

- ✓ Manually configured by a network administrator.
- ✓ The mapping remains fixed unless manually changed.
- ✓ Suitable for critical devices like servers, routers, and printers.
- ✓ More secure but harder to manage in large networks.

Example: Static ARP Entry (IPv4)

On a Cisco router : arp 192.168.1.10 00:1A:2B:3C:4D:5E arpa

On Windows : arp -s 192.168.1.10 00-1A-2B-3C-4D-5E

This ensures that a specific MAC address is always associated with a particular IP.

2. Dynamic Mapping :

- ✓ In **dynamic address mapping**, the mapping between an IP address and a MAC address is done automatically using a protocol. The device learns and updates the mapping as needed.

Characteristics of Dynamic Mapping:

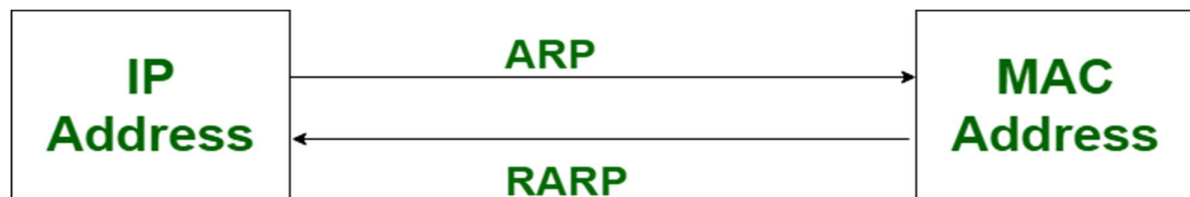
- ✓ Address mapping is done automatically.
- ✓ The system updates the mapping whenever needed.
- ✓ Easier to manage in large networks.
- ✓ Susceptible to spoofing attacks (e.g., ARP poisoning).

Comparison Table: Static vs. Dynamic Mapping

Static Mapping	Dynamic Mapping
Manually assigns IP-MAC mappings.	Automatically assigns mappings using protocols.
Requires manual setup by an admin.	Managed automatically by a DHCP server or ARP.
Less flexible, needs manual changes.	More flexible, updates dynamically.
More secure (fixed addresses prevent spoofing).	Less secure (dynamic addresses can be spoofed).
Harder to manage for large networks.	Easier to manage, especially in large networks.
Used in critical systems (servers, printers).	Used for general-purpose devices (PCs, mobile).

Types of Address Mapping Techniques:

Address mapping techniques are used to translate between logical (IP) and physical (MAC) addresses.



There are **four primary techniques** used in networking for address resolution and mapping:

1. ARP (Address Resolution Protocol) – IP to MAC Mapping

- ARP is Used in IPv4 networks to map an IP address to a MAC address.
- When a device knows the destination's IP address but not its MAC address, it uses ARP to retrieve it
- Command to view ARP table : **arp -a**

Process:

1. A device broadcasts an **ARP request** asking, "Who has this IP address?"
2. The device with the matching IP replies with its **MAC address**.
3. The sender updates its ARP cache for future use.

Example:

- If a computer wants to send data to 192.168.1.2, but only has its IP, it uses ARP to find the MAC address.

Advantages:

- Resolves IP addresses to MAC addresses automatically.
- Reduces manual configuration effort.
- Ensures smooth communication within a local network.
- Stores resolved addresses in a cache for faster access.
- Works efficiently in small to medium-sized networks.

Disadvantages:

- **Vulnerable to security threats like ARP spoofing/poisoning.**
- **Increases network traffic with frequent ARP requests.**
- **Can cause delays if the ARP cache is outdated.**
- **Inefficient in large networks with frequent address changes.**
- **Broadcast-based, which may lead to congestion in busy networks.**

2. RARP (Reverse Address Resolution Protocol) – MAC to IP Mapping

What is RARP?

- ✓ The **Reverse Address Resolution Protocol (RARP)** is a network protocol used to map a **MAC address** to an **IP address**. (opposite of ARP).
- ✓ It is the reverse of **ARP (Address Resolution Protocol)**, which maps an IP address to a MAC address.
- ✓ The device sends a **RARP request** to a RARP server, which responds with an assigned IP address.
- ✓ Rarely used today, replaced by **DHCP (Dynamic Host Configuration Protocol)**.

Why is RARP Needed?

- Some devices, like **diskless workstations**, do not have storage to store an IP address.
- These devices only have a **MAC address** (hardcoded in the network interface).
- They need an **IP address** to communicate on a network.
- RARP helps such devices get an IP address dynamically from a RARP server.

How RARP Works?

- A diskless device knows only its **MAC address**.
- It sends a **RARP request** to a RARP server, asking for its IP address.
- The RARP server looks up the MAC address and responds with the assigned **IP address**.

Key Features of RARP

- ✓ Works at the Data Link Layer (Layer 2)
- ✓ Uses MAC addresses for identification
- ✓ Relies on a dedicated RARP server
- ✓ Does not support subnet masks or additional configurations

◆ Limitations:

- Requires a **RARP server** to function.
- Does not support modern **subnetting** or large-scale networks.
- Replaced by **DHCP (Dynamic Host Configuration Protocol)**.
- **Limited Scalability** – Not efficient for large networks.

❖ Internet Control Message Protocol (ICMP)

What is ICMP?

- ✓ The **Internet Control Message Protocol (ICMP)** is a network layer protocol used for error reporting, diagnostics, and network connectivity troubleshooting in **IPv4 and IPv6 networks**.
- ✓ It works alongside the **Internet Protocol (IP)** to provide feedback about network issues, but it **does not transport application data**.

Key Features of ICMP

- ✓ Used for **error reporting, diagnostics, and operational control**. (unreachable host, packet loss).
- ✓ Does not carry actual data like TCP or UDP.
- ✓ Operates at the network layer (Layer 3) of the OSI model.
- ✓ Works with both IPv4 (ICMPv4) and IPv6 (ICMPv6).
- ✓ Commonly used in **ping** and **traceroute** utilities.

Functions of ICMP :

ICMP mainly performs two functions:

1. **Error Reporting** – Informs the sender when data packets fail to reach their destination.
2. **Network Diagnostics** – Used in tools like **ping** and **traceroute** to check connectivity.

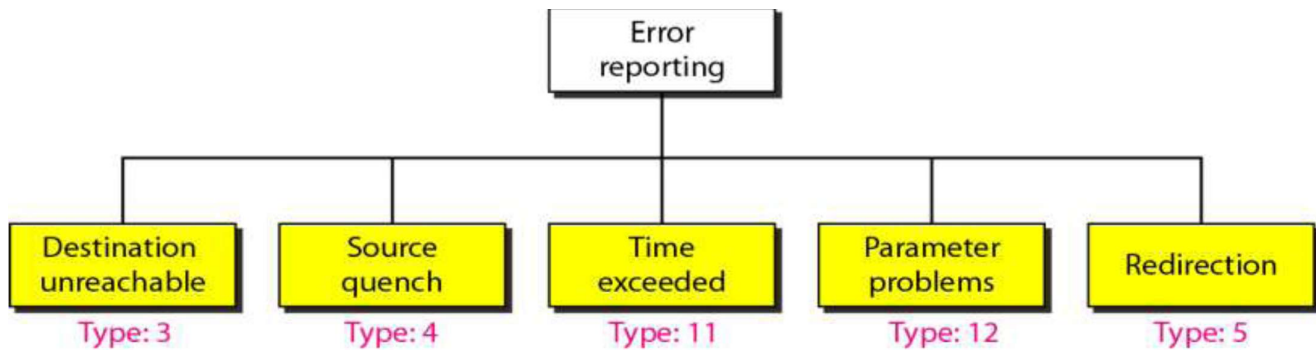
Note :- **ICMP** Always reports error message to the original source

ICMP Message Types : ICMP messages are classified into two categories:

1. **ICMP Error Messages** (indicate network issues).
2. **Query Messages** (used for diagnostics and communication).

1. ICMP Error Messages

ICMP generates error messages when there is a failure in packet delivery. Some common error messages include:



These messages notify the sender about network issues.

ICMP Error Message	Type Code	Description
Destination Unreachable	3	Sent when a destination cannot be reached (host down, network issue).
Time Exceeded	11	Sent when TTL (Time-to-Live) expires in transit (used in traceroute).
Source Quench (Deprecated)	4	Sent to slow down packet transmission to avoid congestion.
Redirect Message	5	Suggests a better route for packet delivery.
Parameter Problem	12	Indicates an issue with the packet header.

2. ICMP Query (Informational) Messages: These messages are used for network diagnostics.

ICMP Query Message	Type Code	Description
Echo Request	8	Sent to check if a host is online (ping command).
Echo Reply	0	Response to an Echo Request (confirms connectivity).
Timestamp Request/Reply	13/14	Synchronizes time between devices.
Address Mask Request/Reply	17/18	Obtains subnet mask information.
Router Solicitation	10	Requests router information (IPv6).
Router Advertisement	9	Provides router information to hosts (IPv6).

Example:

When you **ping** a website, your system sends an **ICMP Echo Request**. If the destination is reachable, it responds with an **ICMP Echo Reply**, confirming network connectivity.

ICMP Advantages & Disadvantages

✓ Advantages

- ✓ **Helps diagnose network problems** (packet loss, high latency).
- ✓ **Provides error reporting** to improve network reliability.
- ✓ **Used in security tools** to detect network vulnerabilities.
- ✓ **ICMPv6 enhances IPv6 performance** by replacing outdated protocols like ARP.

✗ Disadvantages

- ✗ **Can be used for cyberattacks** (ICMP Flood, Ping of Death).
- ✗ **Blocked by some firewalls**, limiting diagnostic capabilities.
- ✗ **Not a transport protocol**, so it does not handle actual data transmission.

BOOTP and DHCP

1. Introduction to BOOTP and DHCP

- ✓ BOOTP (Bootstrap Protocol) and DHCP (Dynamic Host Configuration Protocol) are two protocols used to assign IP addresses dynamically to network devices.
- ✓ DHCP is an advanced version of BOOTP with additional features.

BOOTP (Bootstrap Protocol)

- BOOTP (Bootstrap Protocol) is a **network protocol** used to assign **IP addresses** and configuration details to **diskless devices**
- It is an improved version of RARP (Reverse Address Resolution Protocol), as it provides more configuration information, including:
 - **IP Address**
 - **Subnet Mask**
 - **Default Gateway**
 - **DNS Server**
 - **Boot Server Address** (for network booting)
- Unlike RARP, **BOOTP provides additional information**, such as the default gateway and DNS.
- Developed in the 1980s for diskless workstations to obtain an IP address from a BOOTP server.
- Works on **UDP (User Datagram Protocol)** at **port 67 (server)** and **port 68 (client)**.
- Uses **static mapping**, meaning the server administrator pre-configures MAC-IP bindings.

✓ Advantages of BOOTP

- ✓ Provides more network configuration than RARP (like default gateway, DNS).
- ✓ Works with diskless devices, making network booting possible.

✗ Disadvantages of BOOTP

- ✗ Uses **static IP assignment**, requiring **manual updates** when changes occur.
- ✗ Does not support **dynamic** IP address allocation like DHCP.

BOOTP: Delivery and Forwarding Mechanism

- ✓ BOOTP is an **older protocol** that assigns **static** IP addresses to devices like **diskless workstations, printers, and embedded systems**.

BOOTP Delivery Process

BOOTP works using a **client-server model**. The process involves four steps:

- ✓ **Client Broadcast Request** – A BOOTP client sends a broadcast message requesting an IP address.
- ✓ **Server Response** – The BOOTP server checks its static table and sends a unicast response.
- ✓ **Client Receives IP** – The client uses the assigned IP address and additional configuration details.
- ✓ **No Renewal Mechanism** – The IP address remains assigned until manually changed.

◆ BOOTP Forwarding Mechanism:

- BOOTP **relies on broadcast messages**, which may not work across different networks.
- Routers **must have BOOTP Relay Agents** to forward BOOTP requests between subnets.
- The IP address assignment is **static and manually configured** in the server.

DHCP (Dynamic Host Configuration Protocol)

What is DHCP?

- **DHCP (Dynamic Host Configuration Protocol)** is an advanced version of BOOTP that **dynamically assigns** IP addresses to devices in a network.
- It is the most **widely used protocol** for **automated** IP address management.
- DHCP replaces both **RARP and BOOTP** by allowing **automatic allocation of IP addresses** instead of **manual assignment**.
- Developed as an enhancement of BOOTP with support for **automatic, dynamic, and manual IP assignments**.
- Also uses **UDP ports 67 and 68**.
- Supports **IP lease expiration** and renewal.
- Provides additional network configuration parameters like **subnet mask, gateway, and DNS servers**.

◆ Example

- When you **connect to Wi-Fi**, your device does not have an IP address.
- The **DHCP server** assigns it one dynamically.

✓ Advantages of DHCP

- ✓ **Automatic IP assignment** reduces manual configuration.
- ✓ Supports **both static and dynamic** IP assignments.
- ✓ Handles **large networks efficiently**.

✗ Disadvantages of DHCP

- ✗ **Temporary IP addresses** (leased), requiring periodic renewal.
- ✗ Can cause **network interruptions** when the lease expires.
- ✗ **Security risks** if unauthorized devices obtain IPs from the DHCP server.

DHCP Delivery Process (DORA Process)

DHCP assigns IP addresses dynamically using the **DORA process**:

- ✓ **Discover** – Client broadcasts a DHCPDISCOVER message to find a DHCP server.
- ✓ **Offer** – DHCP server responds with a DHCPOFFER message containing an available IP.
- ✓ **Request** – Client sends a DHCPREQUEST message to confirm acceptance of the offered IP.
- ✓ **Acknowledge** – Server sends a DHCPACK message confirming the lease assignment.

◆ DHCP Forwarding Mechanism:

- DHCP uses **both broadcast and unicast messages**.
- A **DHCP Relay Agent** (router or switch) **forwards requests** to remote DHCP servers.
- Unlike BOOTP, **IP addresses are assigned dynamically**, reducing manual work.

◆ Types of DHCP Address Allocation

1. Static Allocation (Manual Allocation)

◆ What is Static Allocation?

- In this method, a **specific MAC address** (device identifier) is assigned a **fixed IP address**.
- The **administrator manually configures** this mapping in the DHCP server.
- The assigned IP **never changes**, even after reboots or reconnections.

2. Dynamic Allocation (Temporary Assignment)

◆ What is Dynamic Allocation?

- The **DHCP server automatically assigns an IP** to a device for a **specific lease time**.
- Once the lease expires, the **IP can be reassigned** to another device if not renewed.
- This method is useful for **client devices like phones, laptops, and IoT devices**.

3. Automatic Allocation (Permanent Assignment from a Pool)

◆ What is Automatic Allocation?

- The DHCP server **assigns an IP address permanently**, but **without manual configuration** (unlike Static Allocation).
- Once assigned, a device **keeps the same IP unless the server is reset**.

Conclusion

- Static Allocation: Used for servers, printers, routers where a fixed IP is required.
- Dynamic Allocation: Used for client devices (phones, laptops) that do not need a fixed IP.
- Automatic Allocation: Used for workstations, printers, where a device always gets the same IP without manual configuration.

BOOTP vs DHCP

BOOTP	DHCP
Static IP allocation	Dynamic IP allocation
No IP lease expiration	IP lease with renewal
Provides only basic config	Provides full network config
Uses only broadcast	Uses broadcast & unicast
Requires BOOTP relay	Uses DHCP relay for scalability
Slower, predefined database	Faster, dynamic allocation
Limited flexibility	Highly flexible
Used in older networks	Used in modern networks

Unicast Routing Protocols

Unicast Routing Protocols

- ✓ Unicast routing protocols are responsible for determining the best path for **sending data from a single source to a single destination** across a network.
- ✓ These protocols enable routers to dynamically **exchange routing information**, update routing tables, and efficiently forward packets in a network.

Types of Unicast Routing Protocols

Unicast routing protocols are classified into three main types:

- A. **Distance Vector Routing Protocols**
- B. **Link-State Routing Protocols**
- C. **Hybrid Routing Protocols**

A. Distance Vector Routing Protocols

- A **Distance Vector Routing Protocol** is a type of network routing protocol that determines the best path for data packets based on **hop count** (number of routers between the source and destination).
- These protocols follow the **Bellman-Ford algorithm**
- Uses hop count as the metric.
- Periodically shares routing tables with neighbors.
- Example protocols:
 - RIP (Routing Information Protocol) – Max hop count = 15.
 - IGRP (Interior Gateway Routing Protocol) – Cisco proprietary.

◇ Examples of Distance Vector Routing Protocols

Protocol	Metric Used	Maximum Hop Count	Update Interval	Convergence Speed	Best For
RIP (Routing Information Protocol)	Hop Count	15	Every 30 sec	Slow	Small Networks
IGRP (Interior Gateway Routing Protocol)	Bandwidth, Delay	255	Every 90 sec	Medium	Cisco Networks

I. RIP (Routing Information Protocol)

- **RIP (Routing Information Protocol)** is One of the oldest routing protocols, based on **hop count** as a metric.
- Maximum hop count **15** (16 means the destination is unreachable).
- Sends full routing table updates **every 30 seconds** to neighboring routers.

✦ Advantages:

- ✓ Simple & Easy to configure and use.
- ✓ Works well for **small networks**.
- ✓ Prevents routing loops with mechanisms like Split Horizon and Route Poisoning

✦ Disadvantages:

- ✗ **Slow convergence** – takes time to recover from network failures.
- ✗ **Not scalable** – limited to **15 hops**.
- ✗ **High bandwidth usage** – broadcasts **entire routing table** every 30 seconds.

II. IGRP (Interior Gateway Routing Protocol) - Cisco Proprietary

- **IGRP (Interior Gateway Routing Protocol)** is a **distance vector routing protocol** developed by **Cisco** as an improvement over **RIP (Routing Information Protocol)**.
- It is designed for use in **large enterprise networks** and overcomes many limitations of RIP, such as the **15-hop limit** and slow convergence.
- Developed by **Cisco** to improve RIP.
- Uses **multiple metrics** (Bandwidth, Delay, Load, Reliability) instead of just hop count.
- Supports **larger networks** (maximum hop count **255**).
- Sends updates **every 90 seconds**.

✦ Advantages:

- ✓ More accurate path selection (uses multiple metrics).
- ✓ Supports **larger networks** than RIP.
- ✓ **Scalability** – Can handle large networks with up to 255 hops.

Disadvantages of IGRP

- ✗ **Cisco Proprietary** – Only works on Cisco routers.
- ✗ **Slower than modern protocols** – OSPF and EIGRP are preferred today.
- ✗ **Higher resource usage** – Requires more CPU and memory than RIP.

◆ Comparison: RIP vs. IGRP

RIP	IGRP
Uses hop count as the metric	Uses bandwidth, delay, load, and reliability as the metric
Maximum hop count is 15	Maximum hop count is 255 (default 100)
Updates every 30 seconds	Updates every 90 seconds
Slower convergence	Faster convergence than RIP
Suitable for small networks	Suitable for medium to large networks
Uses Bellman-Ford algorithm	Uses Enhanced Bellman-Ford algorithm
No support for multiple metrics	Supports multiple metrics for better path selection
Low CPU and memory usage	Higher CPU and memory usage than RIP
Open standard, works on all routers	Cisco proprietary, works only on Cisco routers

B. Link-State Routing Protocols

What is a Link-State Routing Protocol?

- ✓ A **Link-State Routing Protocol** is a type of **dynamic routing protocol** that builds a **complete network topology** before making routing decisions.
- ✓ Unlike **Distance Vector Routing Protocols**, which rely on periodic updates and hop counts, Link-State protocols use **link-state advertisements (LSAs)** to update the entire network efficiently.

◆ Key Features:

- ✓ Builds a **topological map** of the network.
- ✓ Uses **Dijkstra's Shortest Path Algorithm**.
- ✓ More **accurate** than Distance Vector Routing.
- ✓ Suitable for **large-scale networks**.

◇ Examples of Link-State Routing Protocols

Protocol	Metric Used	Algorithm	Best For
OSPF (Open Shortest Path First)	Cost (Bandwidth)	Dijkstra's SPF	Enterprise Networks
IS-IS (Intermediate System to Intermediate System)	Cost (Configurable)	Dijkstra's SPF	Large ISP Networks

I. OSPF (Open Shortest Path First)

- OSPF (Open Shortest Path First) is a **Link-State Routing Protocol** that dynamically calculates the best path for routing data packets within an IP network.
- It is widely used in large enterprise networks due to its **scalability, fast convergence, and efficient routing**.
- Standardized **open** protocol (not vendor-specific).
- Uses **Cost metric** (based on bandwidth).
- Supports **VLSM (Variable Length Subnet Masking)** and **CIDR (Classless Inter-Domain Routing)**.
- Divides large networks into **Areas** to improve scalability.

✦ Advantages:

- ✓ **Fast Convergence** – Only sends updates when topology changes.
- ✓ **Scalable** – Supports large enterprise networks.
- ✓ **Prevents Routing Loops** – Uses SPF algorithm.
- ✓ **Uses Multicast Instead of Broadcast** – Reduces bandwidth usage.

✦ Disadvantages:

- ✗ **More Complex** – Requires advanced configuration and maintenance.
- ✗ **Higher Resource Usage** – Requires more CPU and memory than RIP.
- ✗ **Requires Proper Network Design** – Poor OSPF area design can cause inefficiency.

OSPF Network Types:

- ◆ **Point-to-Point (P2P)**: Direct connection between two routers.
- ◆ **Broadcast Multi-Access**: Used in networks with multiple routers (e.g., Ethernet).
- ◆ **Non-Broadcast Multi-Access (NBMA)**: Requires manual neighbor configuration (e.g., Frame Relay).

IS-IS (Intermediate System to Intermediate System)

- IS-IS (**Intermediate System to Intermediate System**) is a **Link-State Routing Protocol** used in large-scale networks, particularly in **Service Provider (ISP) and carrier networks**.
- It is similar to **OSPF** but designed primarily for **ISO-based networks** and later adapted for **IP routing** (Integrated IS-IS).
- Developed for **ISO/OSI networks**, but widely used in **ISP networks**.
- Uses a **hierarchical network design** with **Levels 1 and 2** instead of OSPF Areas.
- Supports **IPv4 and IPv6 natively**.

✦ Advantages:

- ✓ **Highly Scalable** – Suitable for **large ISP networks**.
- ✓ **More Flexible** – Allows manual metric configuration.
- ✓ **Supports Fast Convergence** – Like OSPF, uses SPF.

✦ Disadvantages:

- ✗ **Less Common in Enterprises** – Mostly used in ISPs.
- ✗ **More Difficult to Configure** – Requires in-depth knowledge.
- ✗ **Not Natively Designed for IP** – Originally developed for ISO/CLNP routing and later adapted for IP networks.

Comparison: OSPF vs. RIP

OSPF	RIP
Link-State Protocol	Distance Vector Protocol
Faster Convergence	Slow Convergence
No Maximum Hop Count	Limited to 15 hops
Suitable for Large Networks	Suitable for Small Networks
Uses Multicast for updates	Uses Broadcast for updates
Requires more CPU & memory	Requires less CPU & memory

◆ Link-State vs. Distance Vector Routing

Distance Vector Routing	Link-State Routing
Uses the Bellman-Ford algorithm	Uses Dijkstra's Shortest Path First (SPF) algorithm
Sends the entire routing table periodically	Sends only link-state updates when changes occur
Determines the best path based on hop count	Determines the best path based on cost (bandwidth, delay, etc.)
Knows only its neighbors' routes	Maintains a full map of the network topology
Slower convergence due to periodic updates	Faster convergence as updates are event-driven
More prone to routing loops	Prevents loops using SPF calculations
Less scalable, suitable for small networks	More scalable, suitable for large networks
Consumes less memory and processing power	Requires more memory and CPU for calculations
Higher network traffic due to frequent updates	Lower overhead as updates occur only on changes
Examples: RIP, IGRP	Examples: OSPF, IS-IS

C. Hybrid Routing Protocols

- Hybrid routing protocols **combine** the features of **Distance Vector** and **Link-State** routing protocols to provide scalability, efficiency, and fast convergence.
- Combines features of distance vector and link-state protocols.

◆ Key Features:

- ✓ Uses **Distance Vector characteristics** for scalability.
- ✓ Uses **Link-State properties** for accuracy.
- ✓ Best for **large networks** where efficiency and accuracy are needed.

◇ Advantages of Hybrid Routing Protocols

- ✓ Fast convergence and reduced routing loops
- ✓ Scales well for large networks
- ✓ More accurate than Distance Vector protocols

◇ Disadvantages of Hybrid Routing Protocols

- ✗ More complex to configure
- ✗ Higher resource consumption than Distance Vector

EIGRP (Enhanced Interior Gateway Routing Protocol)

- EIGRP (Enhanced Interior Gateway Routing Protocol) is an **advanced distance-vector** routing protocol developed by **Cisco**.
- It improves upon **IGRP** by incorporating features of both distance-vector and link-state protocols, making it a **hybrid routing protocol**.
- **Cisco Proprietary** protocol that combines **Distance Vector and Link-State features**.
- Uses **DUAL (Diffusing Update Algorithm)** for fast convergence.
- Supports **VLSM** and **CIDR**.

✦ Advantages:

- ✓ Fast **convergence** and **reduced routing loops**.
- ✓ Scales well for **large networks**.
- ✓ More **accurate** than Distance Vector protocols.

✦ Disadvantages:

- ✗ **More complex to configure**.
- ✗ **Higher resource consumption** than Distance Vector protocols.

comparison of **RIP, OSPF, and EIGRP** :

RIP	OSPF	EIGRP
Distance-vector protocol	Link-state protocol	Hybrid protocol
Uses hop count as metric	Uses cost (bandwidth) as metric	Uses composite metric (bandwidth, delay, etc.)
Bellman-Ford algorithm	Dijkstra's SPF algorithm	DUAL algorithm
Administrative distance: 120	Administrative distance: 110	Administrative distance: 90 (internal), 170 (external)
Updates every 30 sec (full table)	Updates when topology changes	Updates when topology changes (partial)
Max hop count: 15	No hop count limit	No hop count limit
Slow convergence	Moderate convergence	Fast convergence
Supports equal-cost load balancing	Supports equal-cost load balancing	Supports equal & unequal-cost load balancing
No hierarchical design	Supports areas (hierarchical)	No hierarchical design
Best for small networks	Best for large networks	Best for large Cisco networks
Open standard	Open standard	Cisco proprietary